

Saathvik Kalepu

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EDUCATION

IIITDM Kurnool

B.Tech in Computer Science & Engineering — CGPA: 8.2 / 10.0

Valley Oak Junior College

Class XII (Intermediate) — 98.8%

Andhra Pradesh, India

Aug. 2023 – Expected May 2027

Hyderabad, India

2021 – 2023

EXPERIENCE

DevOps Engineer & UI/UX Designer

Apr. 2026 – Jun. 2026

Caterworld Technologies Pvt. Limited

Hybrid, Chennai

- Deployed and maintained CI/CD pipelines for Progressive Web Application (PWA) modules, streamlining release workflows and reducing deployment cycle time.
- Built responsive, cross-browser UI components and led product iteration cycles, improving front-end performance and feature delivery speed.

AIMaster.live

Hyderabad, India

Software Engineering Intern — Backend & Full Stack

Jul. 2025 – Oct. 2025

- Engineered **20+ RESTful backend API endpoints** using Python/Flask and Node.js/Express.js covering user authentication, session management, database operations, and server-side business logic.
- Optimized database schemas and queries in PostgreSQL/MongoDB, achieving **15% reduction in data retrieval latency** through indexing strategies and query restructuring.
- Built Python/pandas data processing pipelines improving backend system efficiency by **20%**; shipped production-ready features using Git, React.js, REST API contracts, and AWS services.

PROJECTS

RAG Document Assistant | Python, FastAPI, LangChain, FAISS, Docker, Pytest

2026

- Architected a production-grade FastAPI backend with hybrid semantic + keyword retrieval (FAISS + LangChain), hallucination detection, and source-cited answers; validated via **16 automated tests** and **11-case benchmark** at 90%+ retrieval accuracy.
- Containerized with multi-stage Docker and deployed to cloud via render.yaml; designed end-to-end system architecture for production handoff.

AutoVista AI — Car Marketplace Backend | Python, FastAPI, MongoDB, JWT Auth, Pytest

2026

- Designed **30+ authenticated REST API endpoints** for vehicle recommendations, filters, and admin analytics load-tested against **1,000+ simulated users**; implemented JWT auth and RBAC.
- Built a 5-year vehicle price forecasting model and EMI calculator; engineered MongoDB aggregation pipelines for high-performance analytical queries.

CogniShield — Behavioral Security Backend | Python, Flask, SQLite, ML, REST APIs

2025

- Developed real-time anomaly detection backend analyzing keystroke timing, mouse dynamics, and process telemetry with ML-based risk scoring and SIEM-style PDF/JSON export APIs.

TruthLens AI — Multimodal Misinformation Detection | FastAPI, MongoDB, NLP, Computer Vision

2026

- Built backend pipelines for text, image, and video inputs integrating NLP and computer vision for fake-news/deepfake detection with credibility scoring and explainable AI outputs.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Backend & Frameworks: FastAPI, Flask, Node.js, Express.js, REST APIs, JWT Authentication, Microservices

Databases: PostgreSQL, MongoDB, SQLite, Firebase, FAISS (Vector Database)

DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions, CI/CD Pipelines, AWS, Linux, Nginx, Pytest, Postman

Design & Prototyping: Figma, UI/UX Design, Wireframing, Responsive Design, Progressive Web Apps (PWA)

CS Concepts: Data Structures, Algorithms, OOP, System Design, API Design Principles

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Azure AI Essentials (Microsoft, 2024) · Python (HackerRank, 2024) · Data Science (DataCamp, 2024)

Competitive Programming: Solved **200+ LeetCode problems** in DSA and Dynamic Programming

Hackathons & Leadership: Smart India Hackathon · Analytics Vidhya DataHack · Technical Club Member, IIITDM Kurnool